

- MULTI-AGENT DUE-DILIGENCE COMMITTEE

The institutional diligence committee, run by **agents**.

Eight specialist agents across two frameworks read the same room, challenge each other, delegate real tasks, and recruit quantitative help — coordinated end-to-end by **Band** as the genuine coordination layer.

5 TypeScript agents

3 Python · LangGraph specialists

Cross-model: Gemini · Claude · GPT

Auditable, deterministic underwriting

• THE PROBLEM

Real-estate diligence is a committee of experts — and the committee is the bottleneck.

A single deal needs title, zoning, environmental, structural, insurance and financial review. Today that means weeks of email, siloed PDFs, and a partner manually reconciling contradictions before capital is committed.

SLOW & SERIAL

3–6 wks

to clear diligence on a mid-market asset — most of it waiting on hand-offs between specialists.

COSTLY MISTAKES

\$100M+

a single missed easement, zoning conflict or remediation liability can swing an IRR underwater.

NO AUDIT TRAIL

Opaque

why was the deal approved? The reasoning lives in inboxes — impossible to replay or defend.

The value

Compress the committee into minutes, surface contradictions *between* experts automatically, and leave a fully replayable record of every finding, delegation and number — defensible to an investment committee.

● THE SOLUTION

A standing committee of agents that actually talk to each other.

How it works

- 01 Read the room.** Every agent pulls the live Band conversation as its primary context before reasoning.
- 02 Challenge.** When two agents disagree (e.g. an easement one found, another didn't), they negotiate it in-thread.
- 03 Delegate.** Any agent can hand any other a concrete task with intent + authority — decided from context.
- 04 Recruit.** The committee pulls in Python quantitative specialists when the deal warrants it.
- 05 Synthesize.** A Deal Director composes a Red/Yellow/Green memo with conditions precedent.

OUTCOME

A defensible deal memo with a full audit trail — in minutes, not weeks.

NOTHING IS HARDCODED

Which agents debate, who delegates to whom, and which specialists get recruited are **all decided from the deal context** — not a fixed script. Real understanding, not a demo rail.

Eight agents. Two frameworks. One room.

TypeScript ·
reasoning

Python ·
quantitative

ARCHIVIST

Document intake → structured property facts & missing-doc gaps.

REGULATORY

Zoning, permits, flood/FEMA, deed restrictions — severity-ranked.

LEGAL RISK

Title defects, easement conflicts, liens, seller-rep gaps.

FINANCIAL

NOI, cap rate, DSCR & a 5-yr IRR — deterministically computed.

SYNTHESIS

Deal Director — composes the R/Y/G memo & conditions precedent.

ENVIRONMENTAL

Contamination risk + Monte-Carlo remediation cost; Phase I call.

CAPEX / CONSTR.

Renovation/conversion cost & schedule risk, simulated.

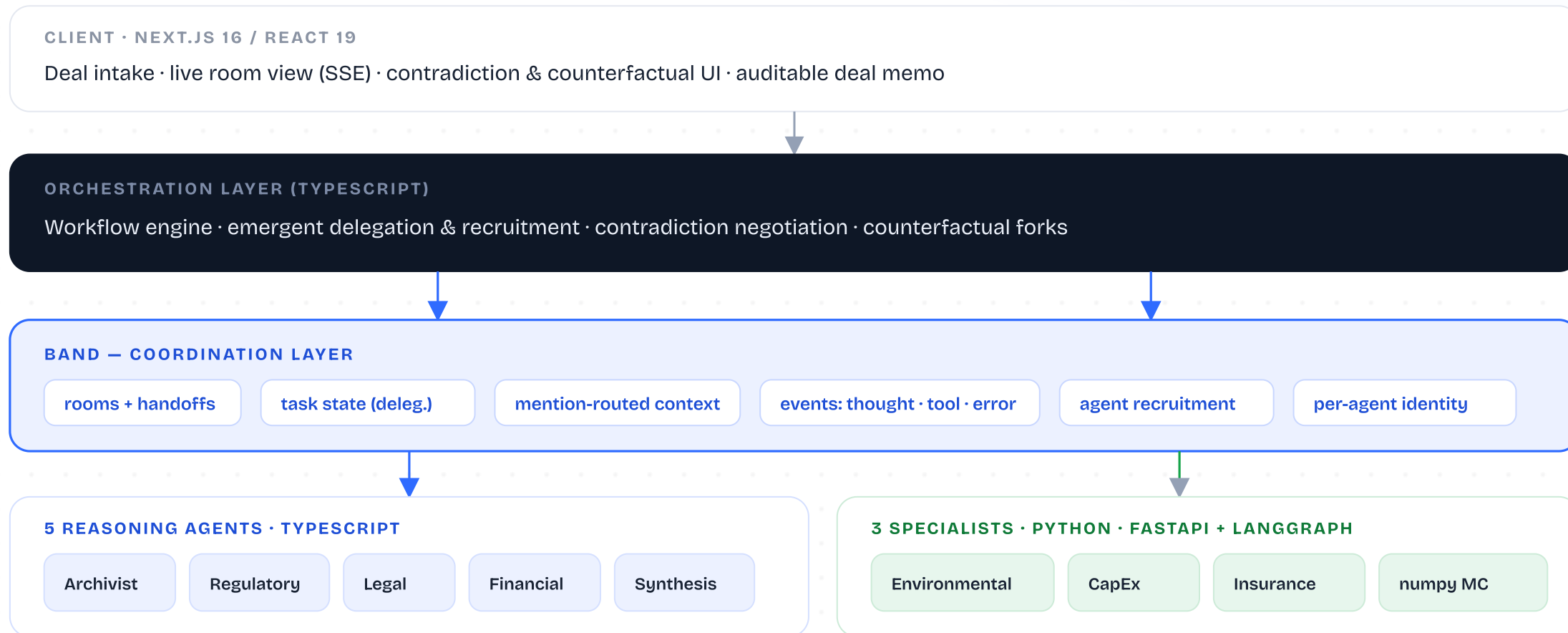
INSURANCE / CAT

Flood/wind/seismic catastrophe exposure & premium estimate.

Why two frameworks?

The 5 reasoning agents live in TypeScript next to the product. The 3 specialists are Python because they do real **numerical work** — seeded numpy Monte-Carlo over LangGraph state machines. Different jobs, right tool each. Band is what makes them one team.

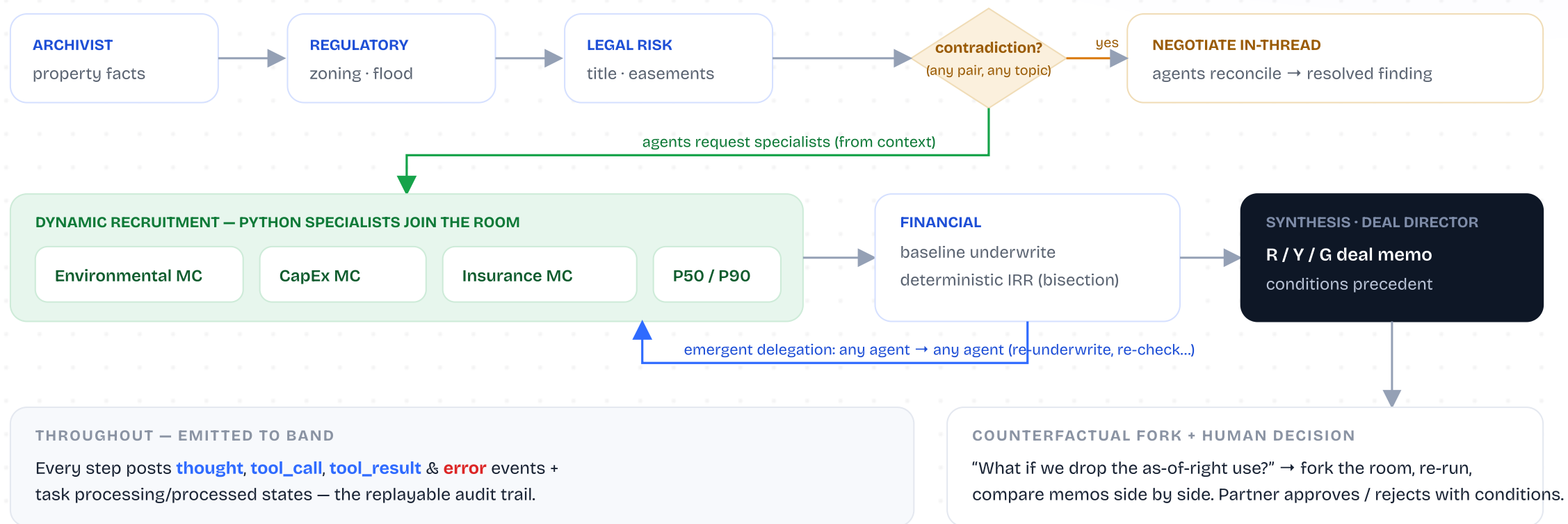
Band is the spine — not a wrapper.



Persistence: Neon Postgres + Drizzle ORM · realtime: SSE bus with replay · models: AI/ML API (Gemini · Claude · GPT)

• WORKFLOW

One deal, end to end.



• WHY THIS NEEDS BAND

We use Band as the coordination layer — to the hilt.

Every primitive below is load-bearing. Remove Band and the committee stops being a committee.

Shared context

Agents **read the room** via mention-routed `getContext` before reasoning — primary context, not spoon-fed payloads.

Handoffs

Structured mention-based posting passes the baton along the committee, each agent addressing the next.

Task state

Delegations are real Band tasks with `processing` → `processed` states — accountable, not fire-and-forget.

Rich events

`thought`, `tool_call`, `tool_result`, `error` — the reasoning is visible and replayable.

Recruitment

Specialists are pulled into the live room mid-deal — cross-framework agents joining as first-class participants.

Distinct identity

Each specialist posts under its **own Band identity** (per-agent key) — genuinely separate agents, not one bot wearing hats.

Four things a wrapper can't do.

01 Contradiction negotiation

When agents disagree — any pair, any topic — they argue it out in-thread until resolved. A deterministic detector guarantees the classic conflicts always fire; an LLM pass discovers novel ones.

02 Emergent delegation

Any agent can hand any other a concrete task — intent + authority — decided from context. Not a fixed cascade: the agents themselves choose who does what next.

03 Dynamic recruitment

The committee decides which Python specialists the deal warrants and pulls them into the room live — cross-framework reinforcements summoned on demand.

04 Counterfactual forking

Fork the room on a “what if”, re-run the committee under a changed assumption, and compare the two memos side by side — diligence as a branchable simulation.

The throughline: every one of these is decided from the deal context at runtime. Nothing is hardcoded — that's what makes it a real conversation between agents, not a scripted demo.

The Python specialists earn their keep.

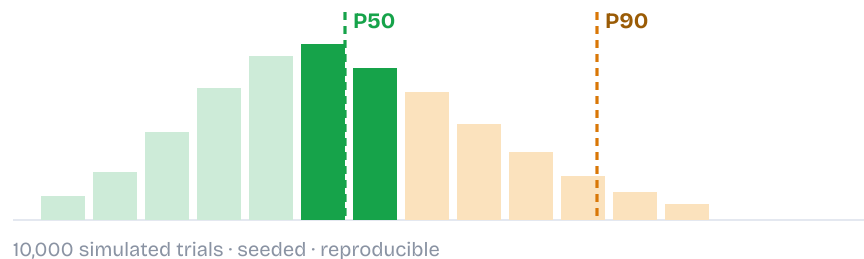
Three LangGraph state machines run seeded numpy Monte-Carlo — genuine quantitative work the reasoning agents can't fake. Recruited live; results posted back to the room.

CAPEX SPECIALIST — LIVE RUN · CEDAR WHARF

\$186M

P50 conversion cost

100% over budget @ P50



ENVIRONMENTAL

Contamination-risk score + remediation-cost distribution → a defensible **Phase I** recommendation.

INSURANCE / CATASTROPHE

Flood/wind/seismic event simulation → expected annual loss & a grounded premium estimate.

WHY PYTHON, HONESTLY

Different framework isn't the point — **value** is. numpy + LangGraph is the right stack for stochastic modelling. Band makes it one committee regardless.

Numbers you can defend to an IC.

DETERMINISTIC UNDERWRITING

The IRR isn't hallucinated. The Financial agent's headline number is computed in code — DSCR and a levered 5-year IRR solved by **bisection** — so the same inputs always produce the same answer.

Baseline IRR

14.2%



After zoning cascade

9.1%

A Critical upstream flag delegates a re-underwrite — and the IRR visibly moves.

FULL AUDIT TRAIL

handoff Archivist → Regulatory: property facts + 2 missing docs

conflict Legal vs Archivist easement — negotiated, resolved

recruit Regulatory pulled in Environmental specialist

task Legal → Financial: re-underwrite · processed ✓

memo Synthesis: **YELLOW** — 3 conditions precedent

Every line persisted to Postgres & replayable — the answer to “why did we approve this?”

Mapped to the judging criteria.

Band coordination

Shared context, handoffs, task-state delegation, rich events, live recruitment, per-agent identity — every primitive is load-bearing, verified live.

Real multi-agent

8 distinct agents that read each other, disagree, delegate and recruit — emergent from context, nothing hardcoded.

Cross-framework

TypeScript reasoning agents + Python/LangGraph quantitative specialists + 3 model families, all in one Band room.

Regulated / high-stakes fit

Deterministic underwriting, severity-ranked findings, conditions precedent, and a complete replayable audit trail.

Polish & reliability

Bounded loops, JSON-repair on every agent, SSE replay, 26 passing tests, clean build — a finished submission, not a prototype.

Built to ship, built to win.

COORDINATION

Band REST — rooms, tasks, events,
recruitment

FRONTEND

Next.js 16 · React 19 · Tailwind 4 ·
SSE

SPECIALISTS

Python · FastAPI · LangGraph ·
numpy

DATA & MODELS

Neon Postgres · Drizzle ·
Gemini/Claude/GPT



**An institutional diligence committee where Band isn't
plumbing —
it's the thing that makes eight agents a team.**

Real coordination. Real cross-framework value. Real, defensible numbers. Nothing hardcoded.